

Derivation of the Line Relationship in a Log-Log Plot

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AST 4762C Homework 4 Question 3.b

Question

A line on a log-log plot has a specific meaning, and is the main purpose of log-log plots. Derive the relationship of x , y , and the slope and intercept of a line in a log-log plot.

Derivation

The main purpose of a log-log plot is to reveal and analyze power-law relationships. A power-law dependence between a dependent variable y and an independent variable x has the general form:

$$y = Ax^m$$

where A is a constant scaling factor (amplitude) and m is the exponent.

To examine how this relationship appears on a logarithmic scale, we take the logarithm of both sides:

$$\log(y) = \log(Ax^m)$$

Using the product rule ($\log(ab) = \log(a) + \log(b)$) and the power rule ($\log(a^b) = b \log(a)$), we expand the expression:

$$\log(y) = \log(A) + m \log(x)$$

This equation matches the standard slope-intercept form of a linear equation, $y = mx + b$, where $y = \log(y)$, $x = \log(x)$, the slope is still m , and $b = \log(A)$.

Therefore, any power-law relation appears as a straight line on a log-log plot. The slope of the line directly gives the exponent m , and the intercept on the vertical axis gives $\log(A)$, allowing the amplitude to be easily recovered as $A = 10^b$ (or e^b if using natural logarithms).